Ink and Paint Analysis

Paint Analysis

Scope of Paint Analysis

- Paint is an **important evidence material**, commonly encountered in cases like motor vehicle accident, theft, burglary or house breaking offences etc.
- Intact paints on houses or machine are also required to be examined at crime scene.
- Evidentiary value of paint increases, whenever it is recovered from a suspect vehicle of a hit and run cases, shoe or clothing of an accused person or from suspected house breaking implements.
- Paints are usually received in the form of flakes, chips, powder or smears.
- Forensic examination of paint can reveal the possibilities and situation under which paint transfer has taken place from one surface to another.
- Comparative study of two or more samples by various techniques can identify whether the samples have originated from a common source.

Paints

- Paint is a coating applied to the surface in form of a liquid dispersion, which is then hardens forming a solid film.

- The functions of paints:

1. Protection of the part surface from the environmental factors (Oxygen and other chemically active gases, moisture, dissolved salts and other chemicals, temperature, bacteria, fungi). Corrosion protection is the most important function of paints.
2. Aesthetic appearance provided by the paint color and sheen (eggshell, satin or gloss).
3. Providing a desired ability of reflection-absorption of heat and light.
4. Changing the surface properties: anti-friction, hardness, electrical conductivity.
5. Identification of products according to the color of the paint.

1. **Paint** - colored non-transparent protective coating.

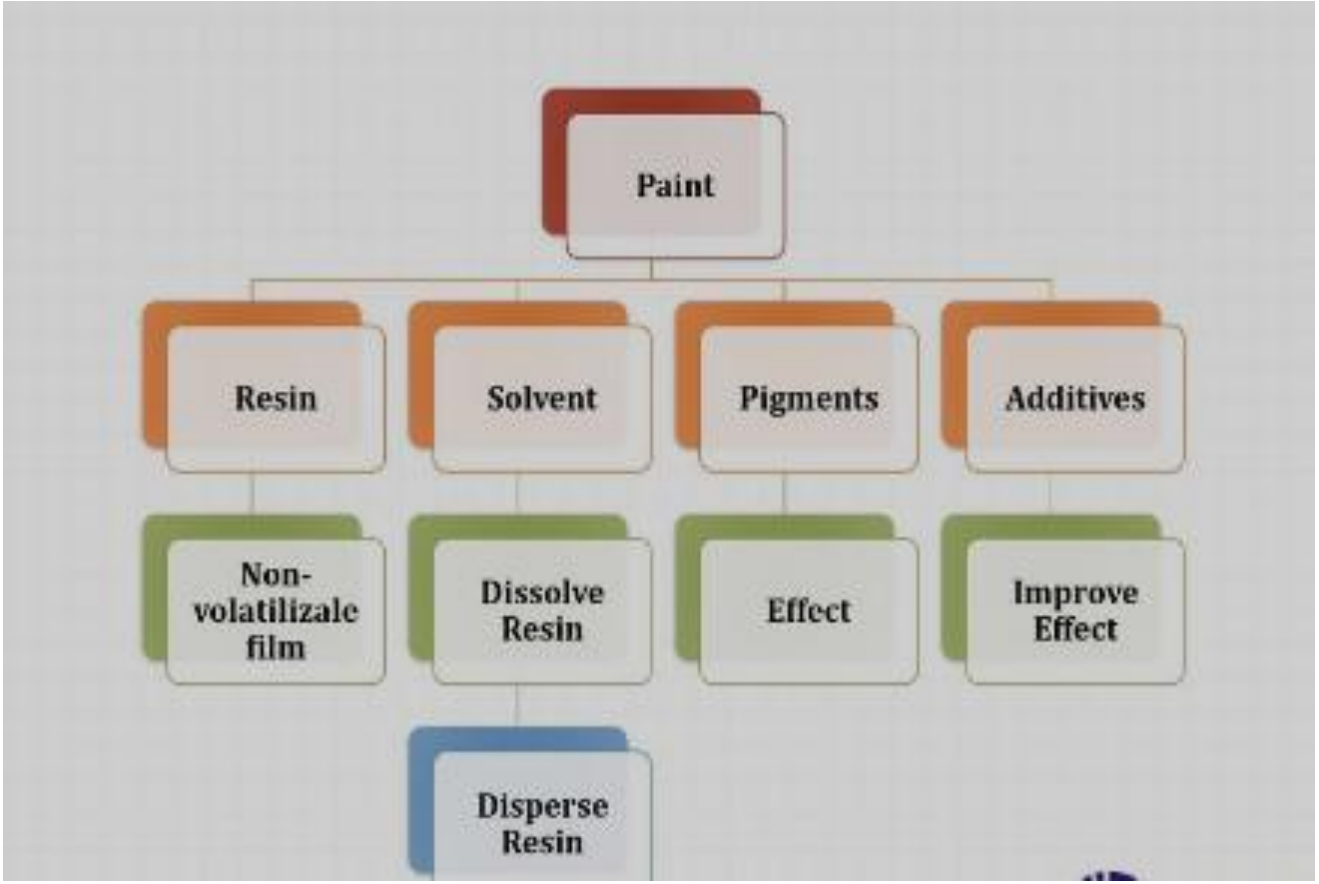
2. **Varnish** - transparent or semi-transparent protective coating.
   - A varnish is made of binder, solvent and additives. Some varnishes contain small amounts of pigment.

3. **Enamel** - hard protective coating with glossy finish.

4. **Primer** - the first coating applied to the surface in order to enhance the adhesion of the final paint (topcoat) and to seal the substrate surface.
   - Primer may be formulated to impart additional protection to the substrate (eg. anti-rust primer for steel substrates).

Paint is composed of

- Binder
- Solvent
- Colorants
- Additives



Binders

- Binder is the main ingredient of paints.
- Binders are polymers (resins) forming a continuous film on the substrate surface.
- Binders are responsible for good adhesion of the coating to the substrate.
- The binder holds the pigment particles distributed throughout the coating.
- The binder is dispersed in a carrier (water or organic solvent either in molecular form (true solutions) or as colloidal dispersions (emulsions or sols).

Common Binders

- The binder is the film-forming component of the paint that holds the pigments and other additives together, creating a solid, continuous film after drying.

- Common binder types include:

1. **Latex (Water-Based):** Used in water-based paints, latex binders are typically acrylic, vinyl acrylic, or other synthetic polymers.
2. **Oil-Based:** These paints use oil-based binders, such as linseed oil or alkyd resins.
3. **Epoxy:** Epoxy resins are used in specialty paints and coatings, known for their durability and resistance to chemicals.
4. **Polyurethane:** Used in high-performance coatings, polyurethane binders offer exceptional durability and UV resistance.

Solvents of Paints

- Solvent (water or organic solvent) is a medium where the binder, pigment and additives are dispersed in molecular form (true solutions) or as colloidal dispersions (emulsions or sols).

- Solvents (thinners) are also used for modification of the paint viscosity required for the application methods: brush, roller, dip, spray.

- The solid coating is formed due to evaporation of solvent

Common Solvents

- Solvents are used to dissolve or disperse the binder and pigments to achieve the desired consistency and application properties.

- The choice of solvent depends on the type of paint:

1. **Water:** Used in water-based (latex) paints.

2. **Mineral Spirits or Turpentine:** Used in oil-based paints.

3. **Volatile Organic Compounds (VOCs):** Solvents in some oil-based and specialty paints can release VOCs into the air, contributing to air pollution.

Colorants

- Colorant is a solid substance dispersed throughout the coating to impart it a color, opacity (hide the substrate surface).
- Colorants may protect the substrate from UV light.
- Colorants change the paint appearance (gloss level) and properties (increase hardness and decrease ductility).
- Colorants may be natural, synthetic, inorganic or organic.
- Fillers and extenders are also referred to pigments.
  - Fillers and extenders are non-expensive commonly natural inorganic materials added to the paint in order to increase its volume, to increase the paint film thickness, to impart toughness or abrasion resistance to the coating.
  - Quartz, talc, limestone etc.

Pigments

- Titanium Dioxide (TiO_2) is a **white** synthetic inorganic pigment existing in two crystalline forms: rutile and anatase. Titanium dioxide has high refractive index (anatase 2.52, rutile 2.76). Anatase is photochemically active but **provides clear white color therefore its main application is interior paints.** Photochemically inert **rutile is used for protection of paints from degradation by light**. Titanium oxide is the most widely used pigment.
- Zinc Oxide (ZnO) is a **white** synthetic inorganic pigment having refractive index 2.01.
- Zinc **Yellow** is Zinc Chromate ($ZnCrO_4$).
- Chrome Oxide Green is olive-green inorganic pigment with a high level of opacity. Chrome Oxide Green is the **most stable green pigment**.
- Phthalocyanine Green imparts **green-blue color**. It is used as the pigment for decorative applications. The pigment is resistant to heat, solvents and alkalis.
- Phthalocyanine Blues are widely spread pigments. They provides a wide spectrum of color: from **reddish-blue to yellowish-green**. The pigments are non-toxic and resistant to solvents, chemicals and elevated temperatures.
- Vermilion is a natural orangish red pigment consisting of toxic mercuric sulfide (HgS).
- Pigment Brown 6 is red inorganic pigment based on Iron(III) oxide (Fe_2O_3).
- Carbon Black is the pigment obtained from organic materials (wood, bones) by charring (thermal decomposition in a limited amount of Oxygen). large quantities of Carbon Black are used for coloring and reinforcing automobile tires.
- Iron (II) Oxide (FeO) is inorganic black pigment.

Additives for Paints

- Additives are small amounts of substances modifying the paint properties.

1. Driers accelerate the paints drying (hardening) by catalyzing the oxidation of the binder.
2. Plasticisers increase the paints flexibility.
3. Fungicides, Biocides and Insecticides prevent growth and attack of fungi, bacteria and insects.
4. Flow control agents improve flow properties.
5. Defoamers prevent formation of air bubbles entrapped in the coatings.
6. Emulsifiers are wetting agents increasing the colloidal stability of the paints in liquid state.
7. UV stabilizers provide stability of the paints under ultra-violet light.
8. Adhesion promoters improve the adhesion of the coating to the substrate.
9. Corrosion inhibitors reduce the corrosion rate of the substrate.
10. Texturizers impart textures to the coatings.

Paint Composition

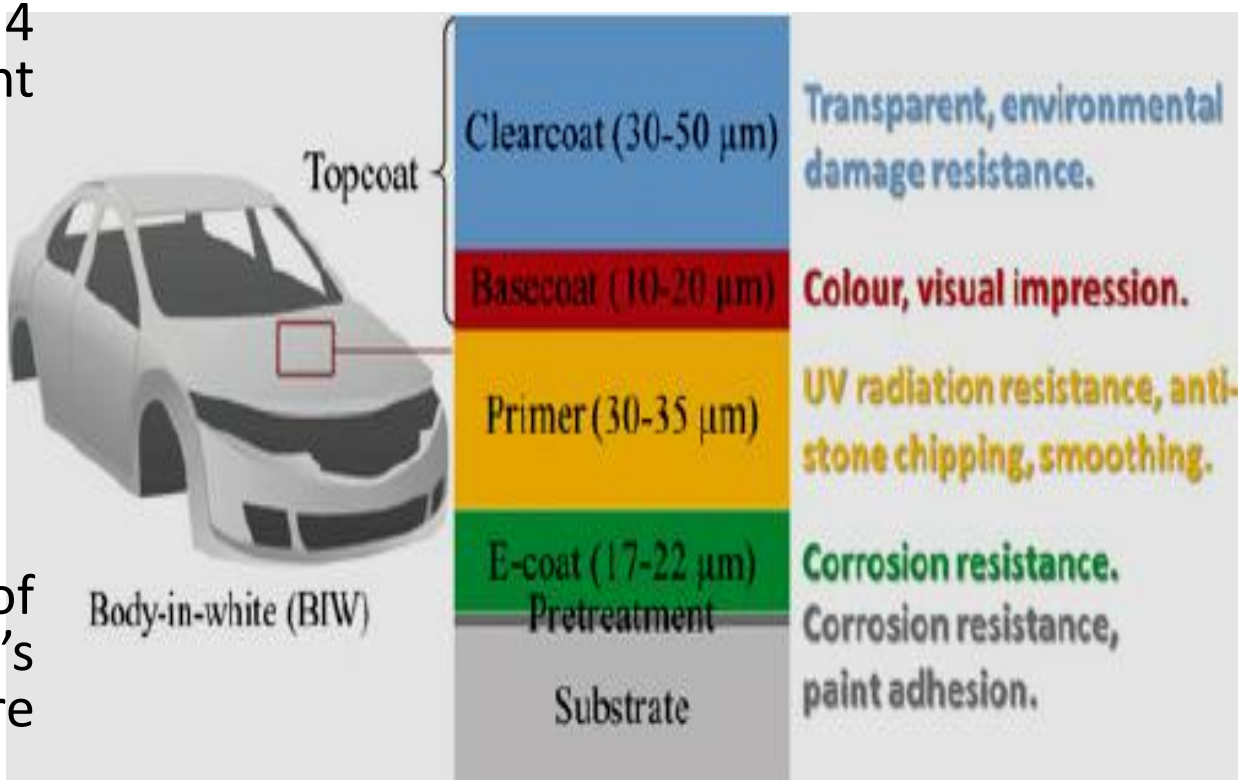
Component	Function
Colorants	Provide color and opacity, and can offer additional protective qualities
Binders (Resins)	Hold the pigment together and help the paint adhere to surfaces, forming a protective film
Solvents (Thinners)	Dissolve and disperse the components of paint, adjusting its viscosity for easier application.
Additives	Enhance specific properties like drying time, durability, and protection against UV light or mold
Fillers	Add bulk, texture, or modify other physical properties of the paint

Automobile Paints

- Automobiles are painted with at least 4 layers of paint formulations.

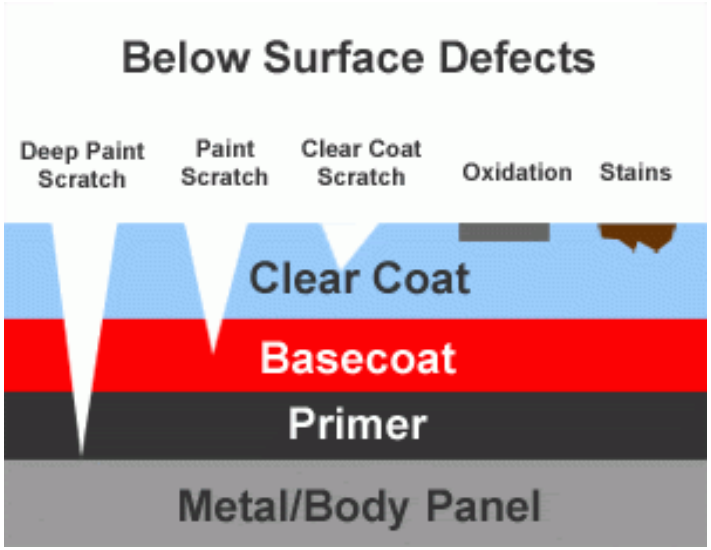
1) Electrocoat primer
2) Primer surfacer
3) Base Coat
4) Clearcoat

- The composition of each manufacturer's paint layers are characteristics.



Significance of multiple layers

- Protection
- Color and aesthetics
- Durability
- Shine and gloss
- UV resistance
- Corrosion resistance
- Chemical resistance
- Value retention
- Quality control
- Brand and image



Electrocoat Primer

- **Sequence (Bottom-most layer)** → First applied directly on the **metal body**.

- **Composition**: Epoxy resins with anti-corrosive pigments.

- **Thickness**: ~20–25 µm.

- **Purpose**
  - Provides **corrosion resistance**.
  - Ensures adhesion of subsequent paint layers.
  - Acts as a protective barrier between metal and environment.

- **Forensic Significance**: Often gray or black in color → provides useful class evidence when compared microscopically.

Primer surfacer

- **Sequence** → Applied above electrocoat primer.

- **Composition**: Polyester resins, epoxy, or urethane-based materials with fillers, talc, and pigments.

- **Thickness**: ~25–35 µm.

- **Purpose**
  - Fills surface imperfections, scratches, or weld marks.
  - Provides smooth surface for basecoat.
  - Contributes to **impact resistance**.

- **Forensic Significance**: Pigment composition can be unique to manufacturer.

Base coat

- **Sequence** → Applied above primer surfacer.

- **Composition**: Acrylic or polyester resins, organic and inorganic pigments, metallic flakes, pearlescent materials.

- **Thickness**: ~10–20 μm.

- **Purpose**
  - Provides the **color and aesthetics** of the vehicle.
  - Metallic/pearlescent pigments give visual effects.

- **Forensic Significance**
  - Most distinctive layer in forensic analysis.
  - Pigments, metallic flakes, and binder ratios can **individualize** a paint chip to a manufacturer/model/year.

Clearcoat

- **Sequence (Top-most layer)** → Final protective coat.
- **Composition**: Urethane or acrylic polymers, UV stabilizers.
- **Thickness**: ~30–50 µm.
- **Purpose**
  - Protects basecoat from UV radiation, weathering, scratches, and chemical exposure.
  - Provides **gloss and depth**.
- **Forensic Significance**: Usually transparent, but presence of specific additives can help in identification.

Differences between Layers

- **Composition:** The primer contains corrosion inhibitors and adhesion promoters, the basecoat contains pigments and binders, and the clearcoat contains UV inhibitors and additives for gloss.
- **Appearance:** The primer is typically matte or satin, the basecoat provides color and finish, and the clearcoat gives gloss and depth.
- **Protection:** The primer protects against corrosion, the basecoat provides color, and the clearcoat provides protection against UV radiation, weathering, and scratches.
- **Thickness:** The primer is the thickest layer, followed by the basecoat, and the clearcoat is the thinnest but the most visible.

Automobile Paint Layers

Primer

- **Purpose:** The primer layer is the first layer applied to the bare metal surface of the vehicle.
- **Function:** It serves as a corrosion-resistant and adhesive layer between the metal substrate and the subsequent layers of paint.
- **Types:** There are different types of primers, such as etch primers for bare metal, primer surfacers for filling imperfections, and epoxy primers for added durability.
- **Color:** Primers are typically gray, black, or white, depending on the type and manufacturer.
- **Thickness:** Primer layers are generally thicker than the other layers to provide proper adhesion and protection.

Base Coat

- **Purpose:** The basecoat is the layer that provides color and appearance to the vehicle.
- **Function:** It is the layer where the pigment is suspended in a carrier or binder.
- **Types:** Basecoats can be solvent-based or water-based, with the latter becoming more common due to environmental regulations.
- **Color:** The basecoat layer determines the color of the vehicle. It can be a single color or a metallic/pearlescent finish.
- **Thickness:** Basecoats are relatively thin compared to the primer and clearcoat layers.

Clear Coat

- **Purpose:** The clearcoat is the final layer applied over the basecoat.
- **Function:** It provides protection, gloss, and depth to the color.
- **Types:** Clearcoats can be high-solids, medium-solids, or low-solids, depending on the desired finish and application method.
- **Color:** Clearcoats are typically transparent or slightly tinted to enhance the color of the basecoat.
- **Thickness:** Clearcoats are relatively thin, but they are the outermost layer and provide the final protective barrier against UV radiation, chemicals, and abrasion.

Microscopic Examination of Paint

- Stereoscopic microscope best
  - Color
  - Surface texture
  - Color layer sequence
  - 10x-125x magnification
  - Not inverted or reversed like regular microscope
  - 3d object can be placed in view
  - Large objects

Pyrolysis

- The decomposition of organic matter by heat
- Solids such as hairs, trace fibers, and paints cannot be dissolved in solvent for normal GC/MS run.
- Materials like these can be heated or pyrolyzed into gas at temperatures of 500-750° Celsius.

Ink Analysis

Scope of Ink Analysis

- Disputed documents/writings
- Counterfeit currency
- Paintings and Reference literature
- Passports/Aadhaar etc
- Graffiti
- Tattoos
- Environmental issues

Ink

- Ink is a type of colored liquid which is used to produce colored surface in different fields like a painting image, text or any kind of design.
- Ink is composed of solvents, pigments, dye, resins, lubricants, solubilizer, surfactant, particular matter, fluoresces and other materials.
- Each component has its own importance to make it more useful.
- However, the Ink formulas vary, but commonly involve four components:
  1. Vehicles (binders)
  2. Colorants/Pigment
  3. Carrier substances/solvent
  4. Additives

Older vs Modern Inks

- The **earliest black writing inks**, developed before 2500BC, were suspensions of carbon, usually lampblack, in water stabilized with a natural gum or materials like egg albumen.
  - The pigment may or may not be colored, and the solvent may be aqueous or organic.
- **Modern ink** formulations are rather more complex.
  - In addition to the pigment, they contain many other ingredients in varying levels. Collectively known as 'vehicle', these additional ingredients include pH modifiers, humectants to retard premature drying, polymeric resins to impart binding and allied properties, defoamer/antifoaming agents to regulate foam efficiency, wetting agents such as surfactants to control surface properties, biocides to inhibit the fungal and bacterial growth that lead to fouling, and thickeners or archeology modifiers to control ink application.

Colorants

- A colorant is essentially any particulate solid - colored, black, white or fluorescent - that alters the appearance of an object by the selective absorption and/or scattering of light.
- Colorants are the chief constituent of an ink and contribute about 50% of its cost.
- It occurs as a colloidal suspension in ink and retains a crystal or particulate structure throughout the coloring or printing process.
- Fluorescent Colorants have a variety of applications, such as in security inks to prevent forgery, in traffic light signals, poster boards and advertising.
- Most commonly used colorants in inks are-
  1. Azo dyes
  2. Copper phathalocyanin blue
  3. Quinacridone
  4. Diaryl pyrrolopyrrole
- The color intensity (strength) of a pigment increases as the particle size reduces

Pigment vs Dye

- **Pigment**
  - A pigment is a dry, powdery substance that must be mixed with a liquid like water to leave behind black, white or color.
  - The pigment is usually water insoluble, but it may remain suspended in the liquid (much as is the case for a “colloid").
  - Pigment particles tend to bond to edges within the chosen medium, a reason why these inks tend to last longer and resist fading over time.

- **Dye**
  - Dye-based inks tend to be cheaper and water soluble.
  - Dye ink generally takes more time to dry and results can be poor if a drop of water hit a page.
  - Colors are brighter because various optical compounds are added to dye ink to enhance the color.
  - However, these compounds have a low resistance to light or to UV rays,
  - Fading can be an issue if your document will be exposed to the sun.

Solvents

- **Function**: Dissolve the dye (or keep the pigment suspended) and adjust the ink's flow properties, drying time, and evaporation rate. The solvent evaporates after application, leaving behind the colorant.

- **Types**:
  1. **Water** (used in water-based inks).
  2. **Oil-based solvents** (used in oil-based inks like those in ballpoint pens).
  3. **Glycols or Alcohols** (used in gel and permanent inks).

Binders and Additives

1. **Binders (Resins or Film Formers)**:
- **Function**: Help the ink adhere to the surface and form a durable film after drying. Binders prevent the ink from smudging and increase its longevity.
- **Types**: Acrylic, shellac, or other polymer-based resins.

2. **Additives**:
- **Function**: Improve specific properties of the ink, such as flow, drying speed, viscosity, or resistance to smearing. They also prevent clogging in pens.
- **Types**:
  - **Surfactants**: Improve wetting and flow properties.
  - **Humectants**: Retain moisture to prevent the ink from drying too quickly inside the pen.
  - **Biocides**: Prevent microbial growth in water-based inks.
  - **Drying Agents**: Help speed up the drying process on paper.

Components of Inks

- Pigments (organic and inorganic)
- Dispersants (surfactants and polymers)
- Resins or polymers improve binding
- Humectants retard premature drying
- Defoamers and antifoaming agents
- Wetting agents enhance contact with the substrate
- pH modifiers (usually amine derivatives)
- Biocides and bacteriostats

Ink Composition

Component	Function	Examples
Colorants (Dyes/Pigments)	Provide color and contrast on paper	Dyes: Crystal Violet, Methylene Blue; Pigments: Carbon Black, Iron Oxide
Solvents	Dissolve the colorants and facilitate flow	Water (for water-based inks), Glycol, Ethanol, Isopropanol
Binders/Resins	Improve adhesion to surfaces and drying properties	Acrylic resins, Shellac, Polyvinyl acetate
Additives	Enhance ink properties (e.g., drying time, stability, pH balance)	Surfactants, Lubricants, Antioxidants, Biocides
Preservatives	Prevent microbial growth in ink	Phenol, Sodium Benzoate

Types of Inks

- **Ballpoint ink** is specially formulated to be thick and fast-drying. Its viscosity is precisely controlled: it must be as thick as possible but still thin enough to flow down the barrel of the pen in response to gravity.
  - Ballpoint pen inks use dye because the tiny particles of undissolved pigment can clog the ball of the pen.
  - Ballpoint pen ink is usually oil-based to give it its thickness. Oil also is why ballpoint pen ink dries quickly and is permanent and water-fast.
- **Fountain pen ink** is a water based ink (dye) intended for use with fountain pens.
  - Red ink usually contains the eosin dye where as blue ink often contains triarylmethane dye.
- **Gel pen ink** is a pigment suspended (colloidal) in water based gel and it is opaque and thick in appearance.

Based on Writing Instruments

Type of Ink	Description	Example Applications
Ballpoint Pen Ink	Thick, oil-based ink containing dyes and resins; dries quickly and forms clumps on paper.	Most commonly used in signatures, legal documents, and everyday writing.
Gel Pen Ink	Water-based ink containing pigments suspended in a gel; provides smooth writing.	Used in artistic applications, personal notes, and decorative writing.
Fountain Pen Ink	Water-based liquid ink; does not contain resins and spreads easily on paper.	Used in calligraphy, official and historical documents.
Rollerball Pen Ink	Water-based ink with fluid flow, similar to fountain pen ink but used in rollerball pens.	Used for smooth and fine writing.
Printer Ink (Inkjet/Toner)	Dye-based (inkjet) or powder-based (laser toner) ink used in printing documents.	Used in printed documents, fraudulent check printing.
Stamp Ink	Dye-based ink used in rubber stamps, official seals, and financial documents.	Used in legal paperwork, passports, and currency authentication.

Based on Special Forensic Characteristics

Type of Ink	Description	Example Applications
Security Ink (UV/IR Reactive Ink)	Fluoresces under ultraviolet (UV) or infrared (IR) light, used for authenticity verification.	Found in currency, passports, and official documents.
Thermochromic Ink	Changes color when exposed to heat.	Used in security printing, tamper-proof documents.
Invisible (Steganographic) Ink	Appears only under UV light or after chemical treatment.	Used in confidential markings and security labels.
Solvent-Sensitive Ink	Reacts with chemicals or erasures, making alterations visible.	Used in fraud prevention in checks and legal documents.
Magnetic Ink	Contains magnetic particles for machine readability.	Used in banknotes and MICR (Magnetic Ink Character Recognition) checks.

Systemic Approach to Forensic Ink Analysis

1. Physical examination

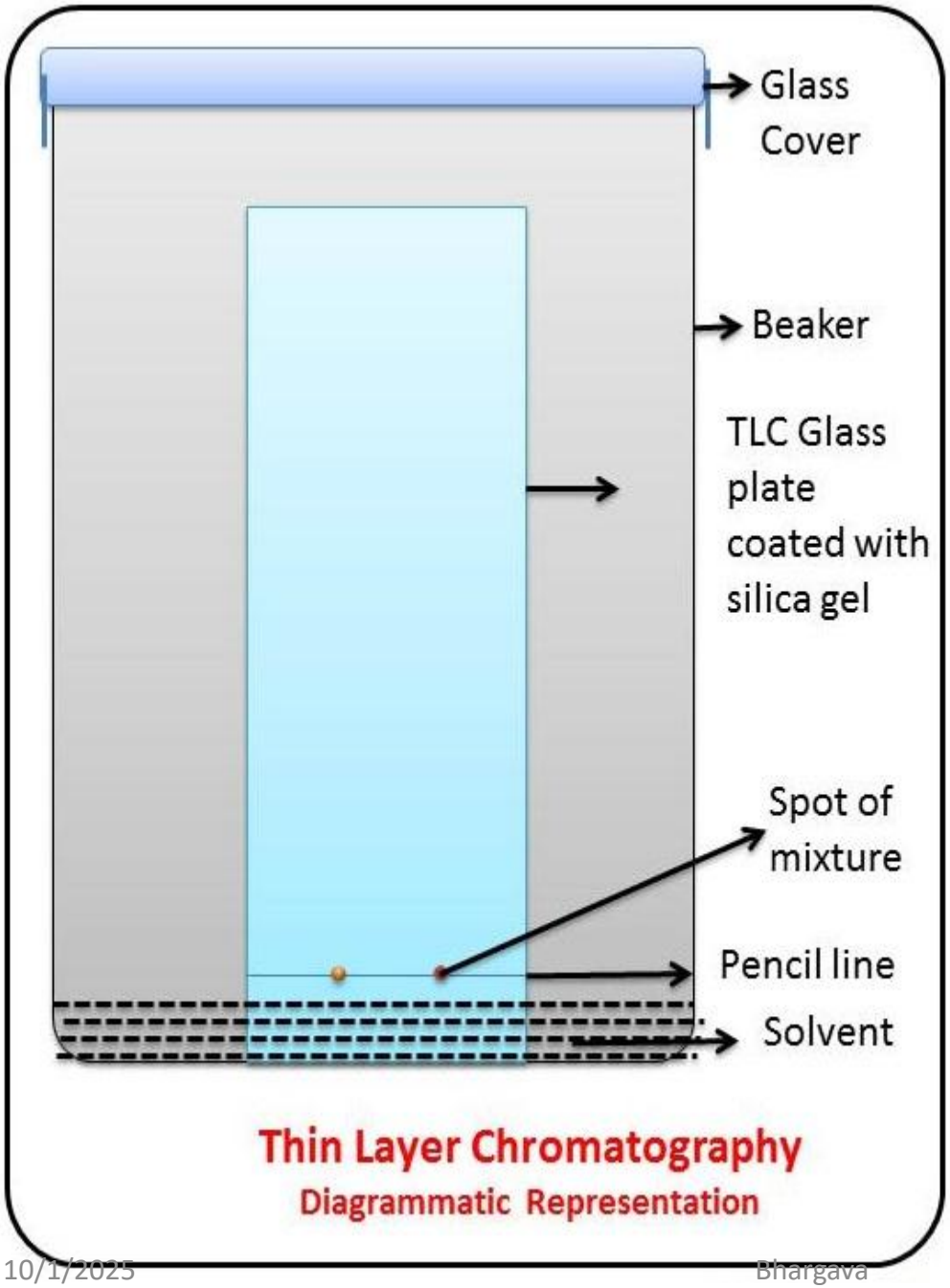
- This includes the optical examination of ink with the help of hand magnifier or compound microscope to determine:
  - The type of ink used (ball point pen, fountain pen or fiber tip etc)
  - Color of ink
  - Comparison of secondary color shades, ultraviolet rays are used to compare the degree of fluorescence.
  - Infrared rays are used to differentiate dyes and pigments and especially ball point pen inks.

2. Chemical examination

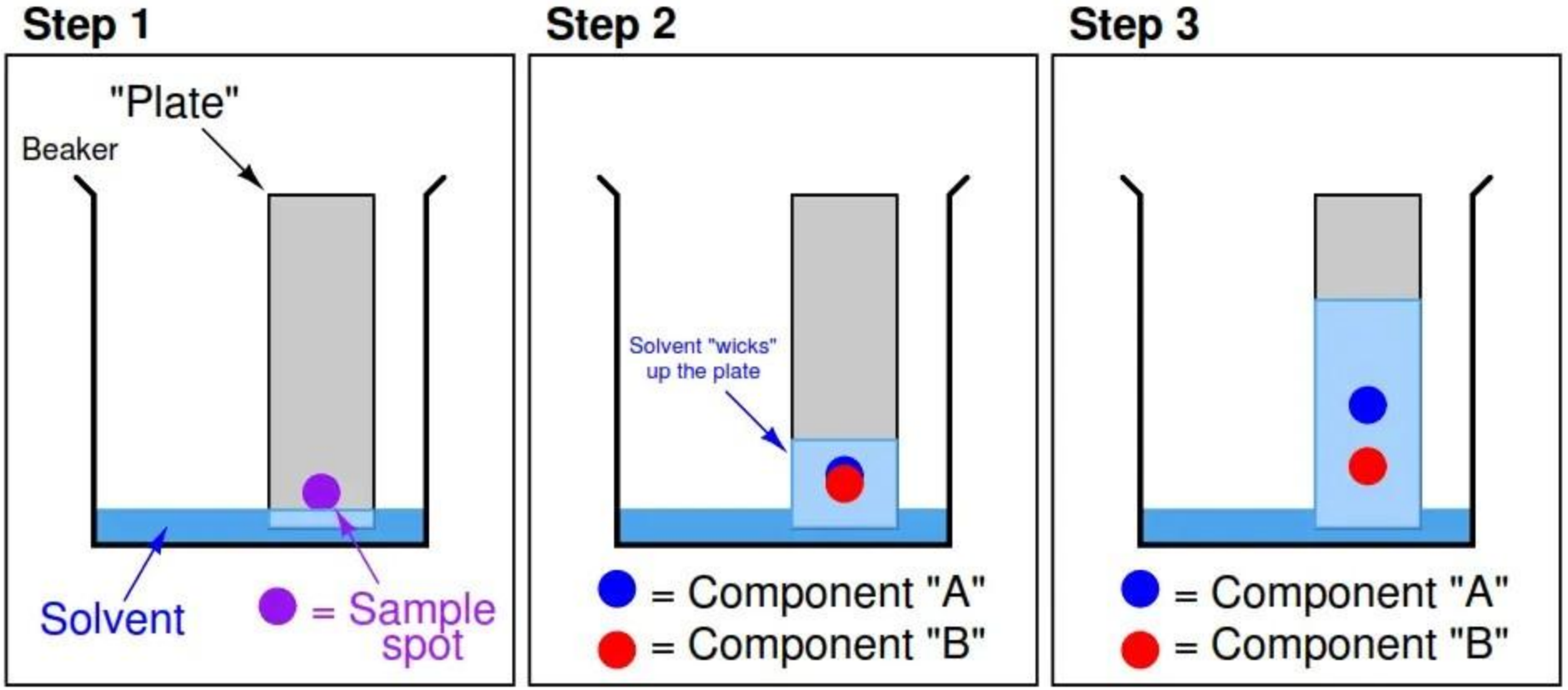
- The chemical analysis of ink can be conducted in two different ways-
  - **Preliminary tests:** Performing chemical spot tests on the punched-out fragments of ink strokes or on the ink strokes itself.
  - **Confirmatory tests:** Chromatographic analysis for isolation and characterizing various dye stuff inks.

Ink and Paint Analysis

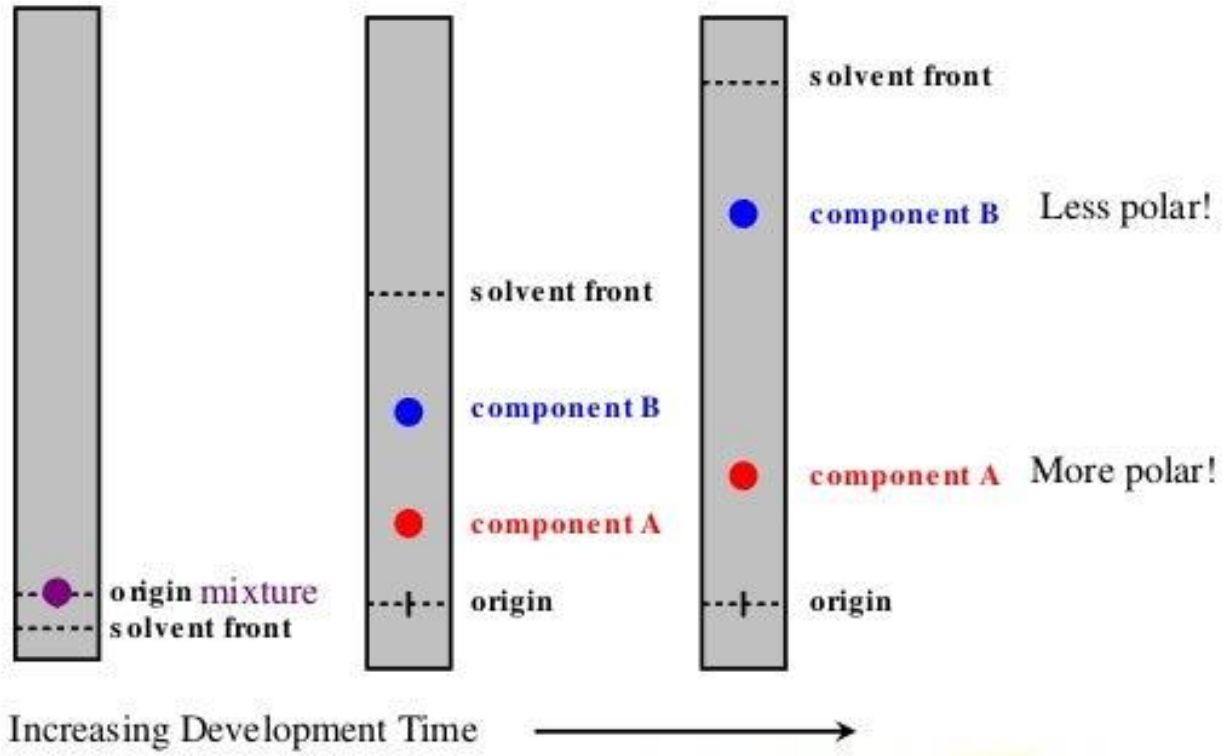
(Thin Layer Chromatography)


Glass
Cover
Beaker
TLC Glass
plate
coated with
silica gel
Spot of
mixture
Pencil line
Solvent
Thin Layer Chromatography
Diagrammatic Representation

Thin-layer chromatography



Thin-Layer Chromatography: A Two-Component Mixture



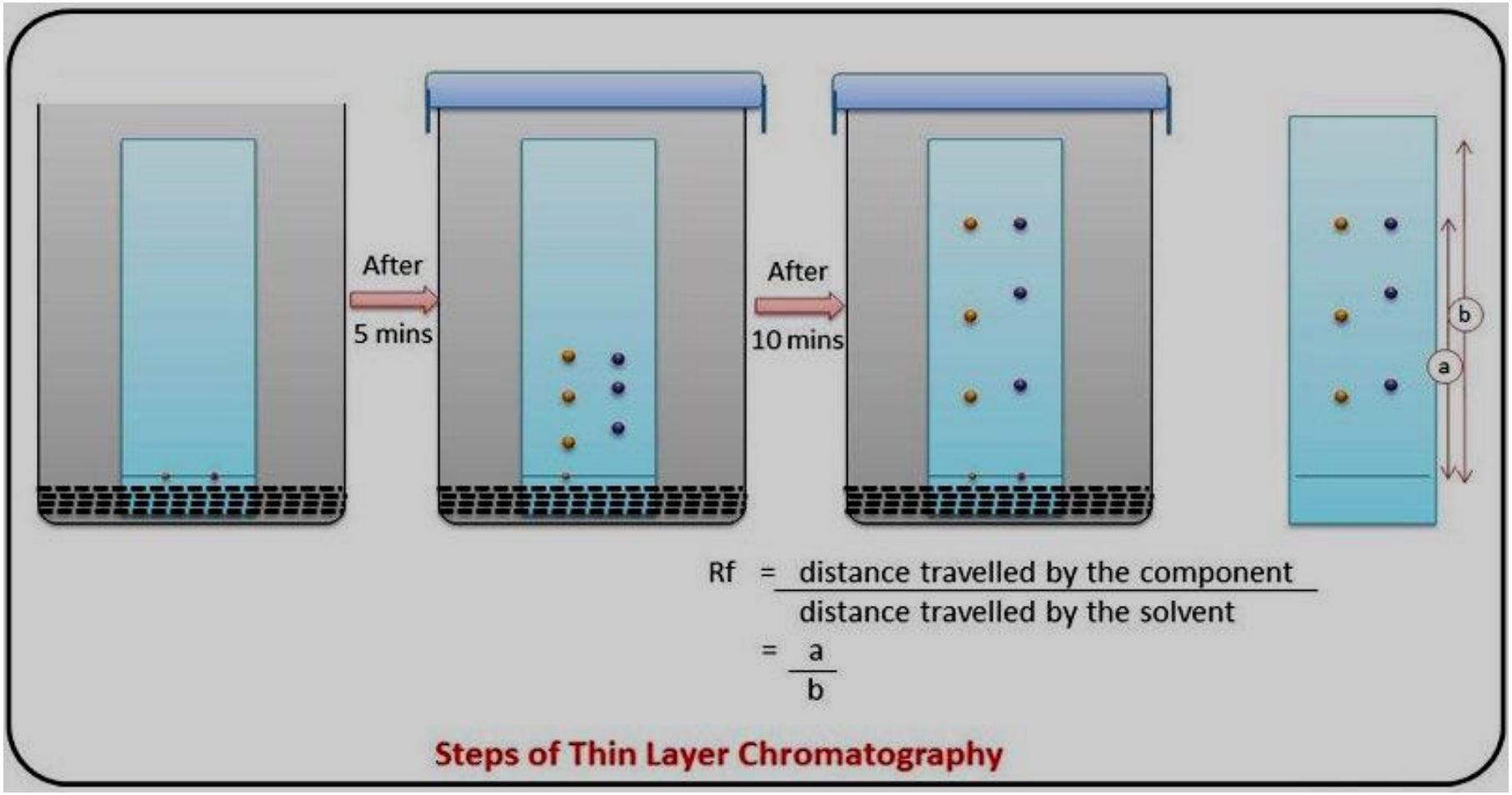
Chromatographic Analysis/TLC of Inks

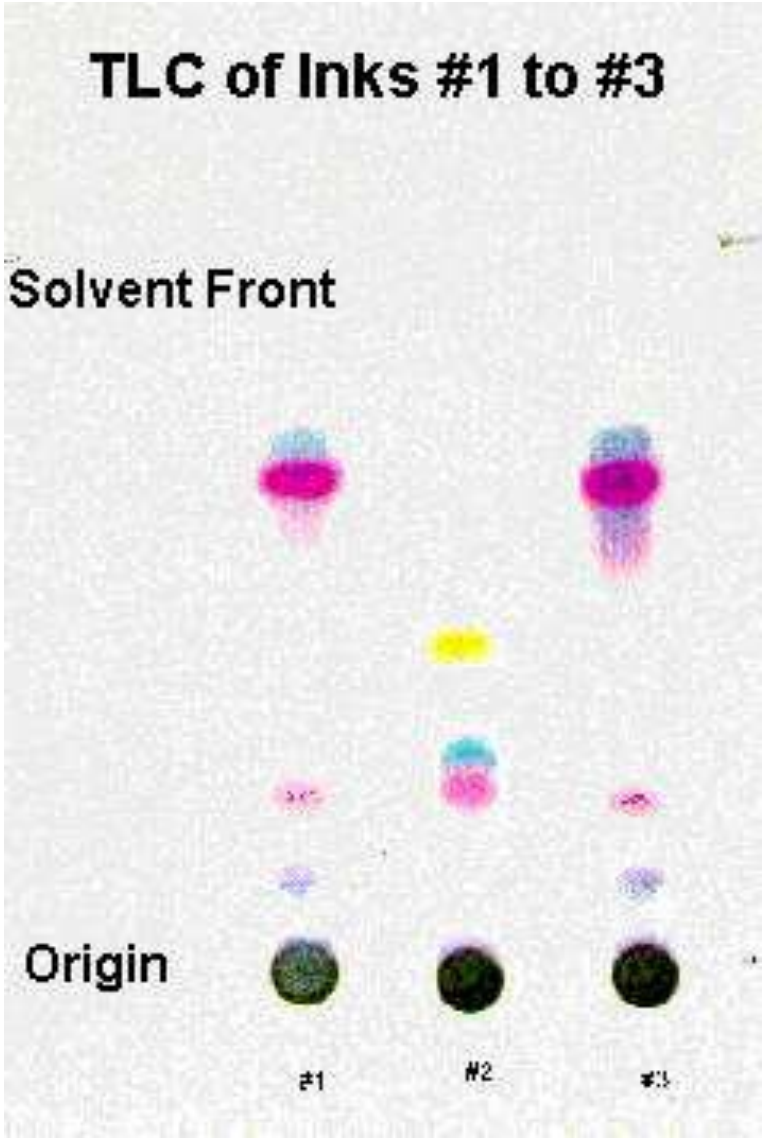
- Principle of chromatography:
  - Distributes of sample into two immiscible phases (stationary and mobile phases) which is described by partition or distribution coefficient, Kd.

- Thin layer chromatography (TLC) is most commonly used methods for separation of ink components and comparisons of various inks.

- For differentiation of two inks, their spots are marked on the same TLC plate and if these samples show same number of spots with the same Rf values and colors, then the two inks are identical in their dyes composition otherwise not.

$$R_f = \frac{\text{Distance travelled by solute in stationary phase}}{\text{Distance travelled by solvent front}}$$


After
5 mins
After
10 mins
b
a
Rf = distance travelled by the component / distance travelled by the solvent
= a / b
Steps of Thin Layer Chromatography


TLC of Inks #1 to #3
Solvent Front
Origin
#1
#2
#3

Commonly used Mobile Phases for Ink Analysis

1. Butanol: ethanol: water – 50:15:10
2. Ethyl acetate: cyclohexane: methanol: ammonia – 70:15:10:5
3. Ethyl acetate: butanol: ammonia – 50:35:5
4. Ethyl acetate: ethanol: water – 70:35:30
5. Toluene: acetate: ethanol: water – 30:60:7:2

Differential Staining Analysis

- Use of specific chemical dyes to differentiate and identify various components in paper.

- In the context of paper analysis, two prominent stains are commonly used:
  1. **Herzberg's stain**
  2. **Jenk's stain**

- These stains help forensic examiners differentiate between the types of fibers present in a paper sample, aiding in the identification of its origin or manufacturing process.

Herzberg's Stain

- Herzberg's stain is a combination of **iodine and zinc chloride**
- Used to distinguish between different types of cellulose fibers commonly found in paper.
- It has different effects on different types of fibers, allowing analysts to differentiate among **cotton, linen, wood pulp,** and other materials.

- When applied to paper:
  1. **Cotton and Linen fibers**: The stain turns these fibers a characteristic **blue or purple** color.
  2. **Wood fibers**: These turn a **yellow or orange** hue under the stain.

- This method is particularly useful in determining whether the paper was made from wood pulp, rag pulp, or a combination of different fibers.
- It's **helpful in identifying counterfeit documents** and understanding the composition of historical or criminally relevant papers.

Jenk’s Stain

- Jenk’s stain is used for distinguishing between hardwood and softwood fibers in paper samples.

- Hardwood fibers tend to stain differently than softwood fibers, making it easier for forensic examiners to determine the paper's source material.
  1. **Hardwood fibers** typically stain **red or pink**.
  2. **Softwood fibers** usually stain **yellow or orange**.

- Jenk’s stain is especially useful for identification of document forgery, counterfeiting, or any investigation where the origin and type of paper are crucial.

Forensic Application of Differential Staining Methods

- In forensic investigations, these stains are part of the microchemical testing procedures used to analyze the paper's fiber composition.

- By determining the type of fiber, its origin, and other microscopic characteristics, forensic scientists can provide key information in criminal cases, including:

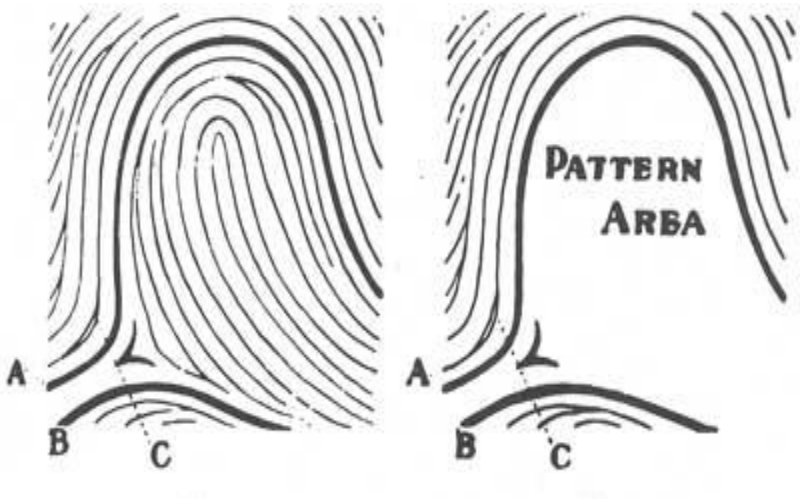
1. **Document authenticity**: Identifying paper used in counterfeit documents, forgeries, or altered legal papers.
2. **Paper tracing**: Linking a document to a specific manufacturer or time period.
3. **Historical document analysis**: Determining the authenticity and origin of ancient documents or records.

Finger Prints in Forensics

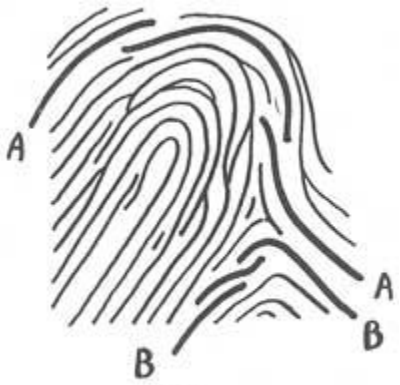
Basics- (Pattern Area, Type Lines, Delta, Core)
Patterns of Finger Prints
Role of Sweat and Sebaceous Glands in FPs
Fingerprint Residue
Types of substrate and Fingerprints
Developing Methods

Elements of FingerPrints

- **Pattern Area:**
  - *The pattern area is that part of a loop or whorl in which appear the cores, deltas, and ridges with which we are concerned in classifying.*
  - The pattern areas of loops and whorls are enclosed by type lines.
- **Type Lines:**
  - *Type lines may be defined as the two innermost ridges which start parallel, diverge, and surround or tend to surround the pattern area.*



Rules governing Type Lines (TL)



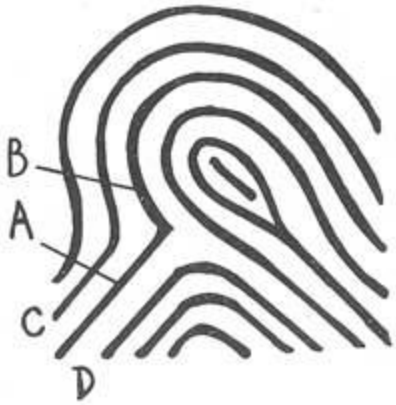
13

Broken Type Line



14

Small Type Line



18

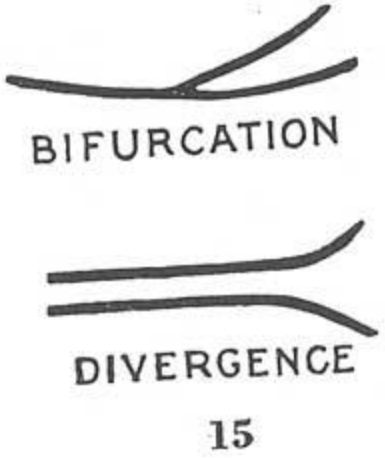
TL must not be angled but curved/diverged



16



17



15

TL must be divergences NOT bifurcation

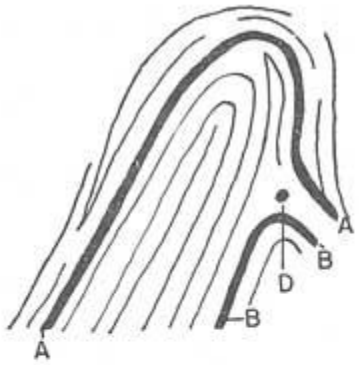
Elements of FP

- **Focal points**
  - Within the pattern areas of loops and whorls are enclosed the focal points which are **used to classify them**. These points are called **delta** and **core**.

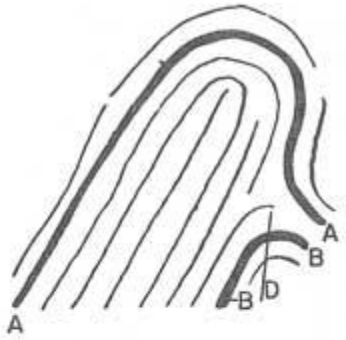
- **Delta**
  - *is that point on a ridge at or in front of and nearest the center of the divergence of the type lines.*

- Delta may be:
  - A bifurcation
  - An abrupt ending ridge
  - A dot
  - A short ridge
  - A meeting of two ridges
  - A point on the first recurving ridge located nearest to the center and in front of the divergence of the type lines.

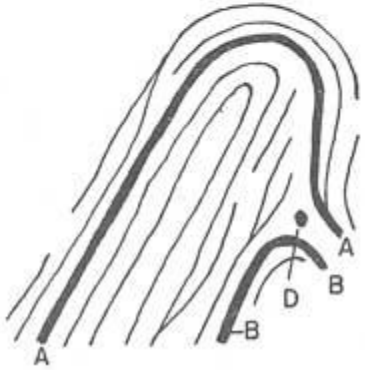
Delta



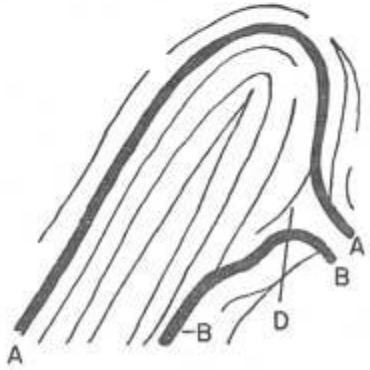
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24

The Core

- **Core**
  - is the approximate center of the finger impression.
  - It will be necessary to concern ourselves with the core of the loop type only.

- The following rules govern the selection of the core of a loop:
  - *The core is placed upon or within the innermost sufficient recurve.*

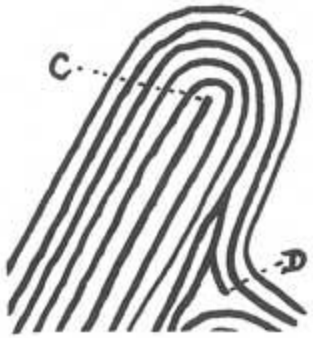
  1. *When the innermost sufficient recurve contains no ending ridge or rod rising as high as the shoulders of the loop, the core is placed on the shoulder of the loop farther from the delta.*
  2. *When the innermost sufficient recurve contains an odd number of rods rising as high as the shoulders, the core is placed upon the end of the center rod whether it touches the looping ridge or not.*
  3. *When the innermost sufficient recurve contains an even number of rods rising as high as the shoulders, the core is placed upon the end of the farther one of the two center rods, the two center rods being treated as though they were connected by a recurving ridge.*

  - *The shoulders of a loop are the points at which the recurving ridge definitely turns inward or curves.*

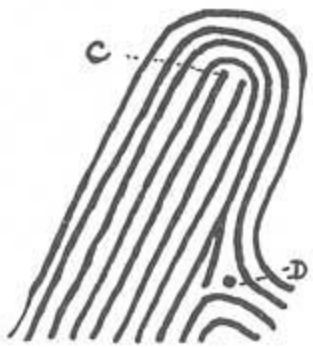
Focal Points



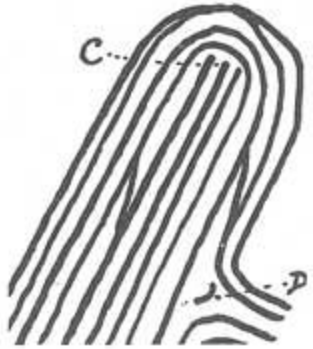
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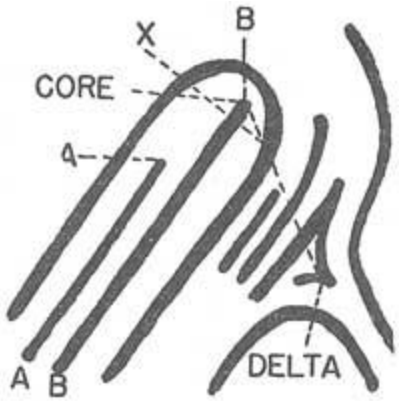
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Finger print patterns

Basic Pattern	Pattern type	Symbol
Arch	Plain Arch	**A**
	Tented Arch	**T**
Loop	Radial Loop	**R**
	Ulnar Loop	**U**
Whorl	Plain Whorl	**W**
	Central Pocket Loop	**C**
	Double Loops	**S**
	Accidental	**X**

Arch

•Arches are found in about 5% of fingerprint patterns encountered.

•In arches, the ridges of the finger run continuous from one side of the finger to the other with no re-curving.

•It is imperative to state that the Arches show absence of the any of the Type Lines, Deltas or Cores.

Plain Arch

- This pattern has a steady flow of ridges.
- It begins from the one side of the finger, and then the ridge flows upward marginally, almost like a wave. The plain arch then continues its trip along the finger to the other side.

Tented Arch

- This pattern is similar to the plain arch; it begins on one side of the finger and glides out in a similar manner to the other side.
- However, the dissimilarity lies in the ridges. In a plain arch, the ridges are continuous while in tented arch these are not. The ridges, which connect each other in the center, meet and thrust upward, giving the imprint the shape of a tent.

Arches

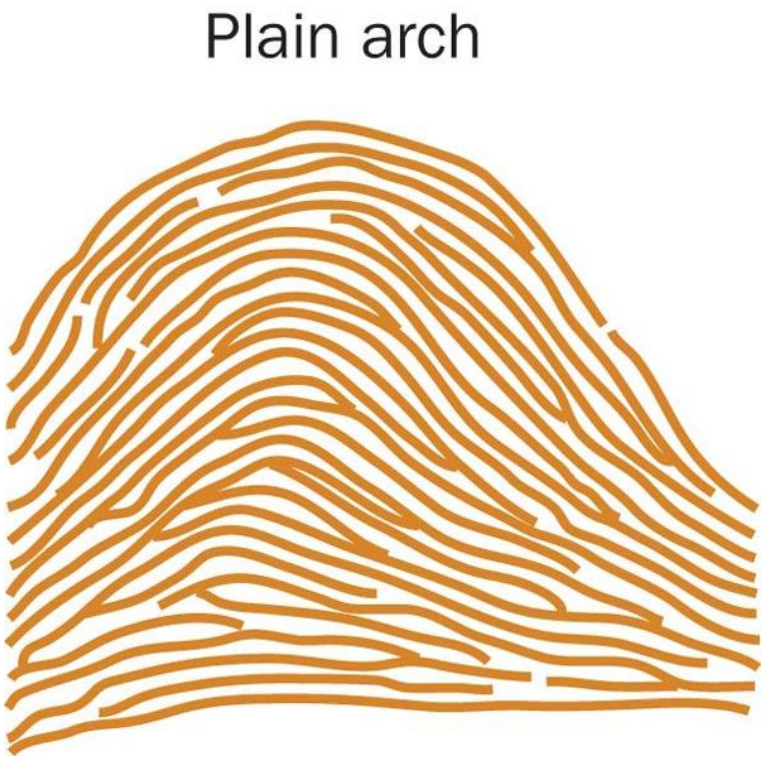
Plain Arch Pattern **(4%)**

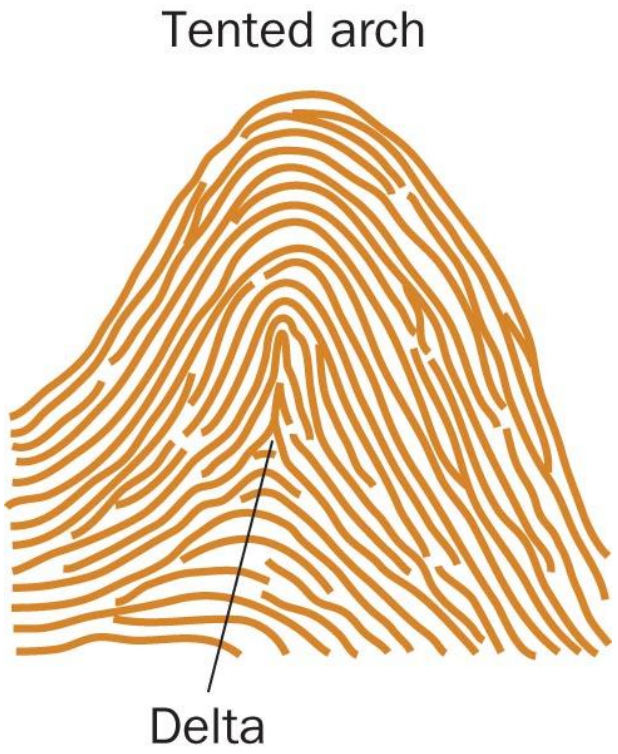


Tented Arch Pattern **(1%)**



Arches





The LOOP

- *A loop is that type of fingerprint pattern in which one or more of the ridges enter on either side of the impression, recurve, touch or pass an imaginary line drawn from the delta to the core, and terminate or tend to terminate on or toward the same side of the impression from whence such ridge or ridges entered.*

- ***Essentials of a loop***
  - A sufficient recurve.
  - A delta.
  - A ridge count across a looping ridge.

- *A sufficient recurve may be defined as that part of a recurving ridge between the shoulders of a loop. It must be free of any appendages abutting upon the outside of the recurve at a right angle.*

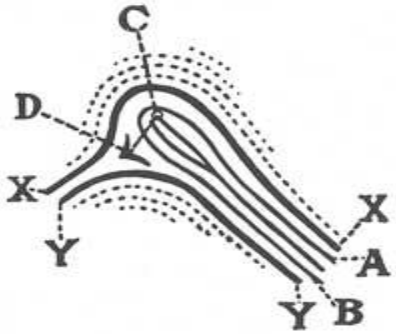
How to define the LOOP



56



57



60



61



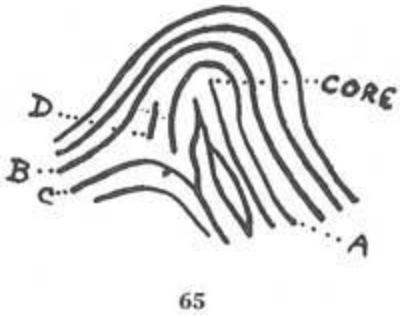
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63



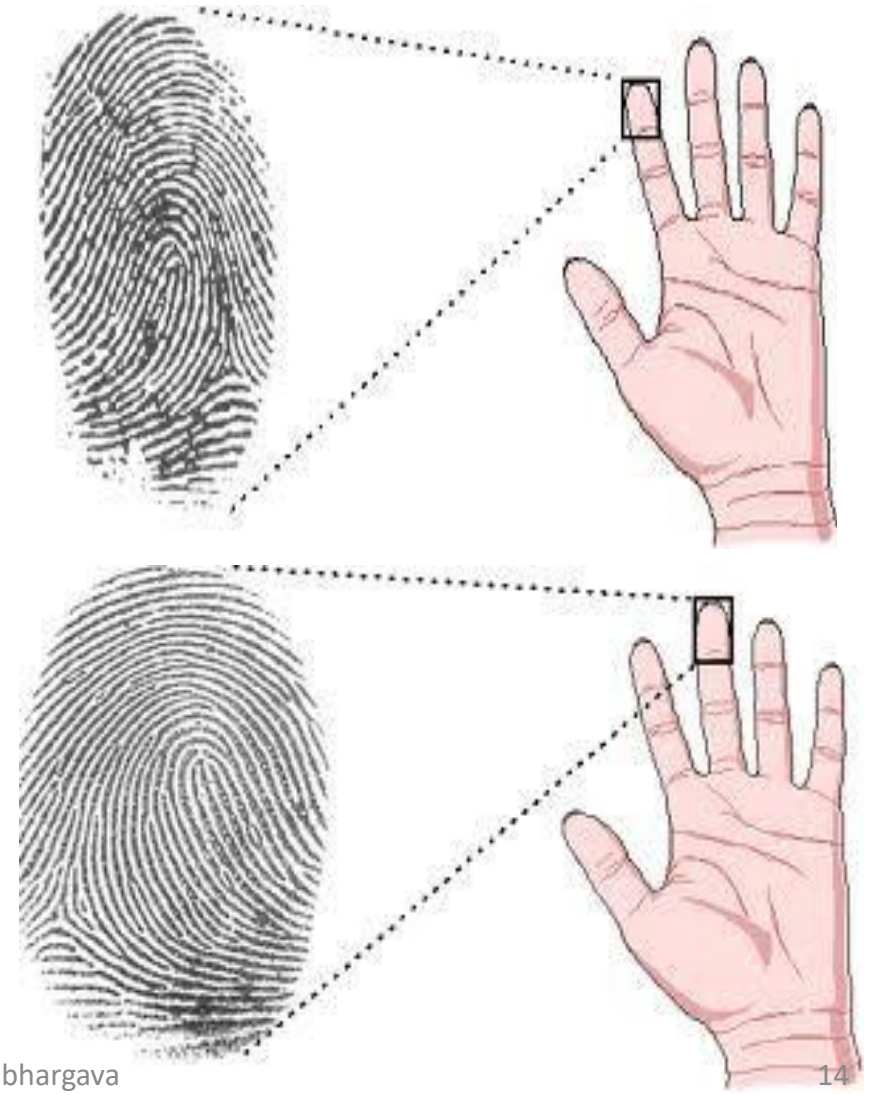
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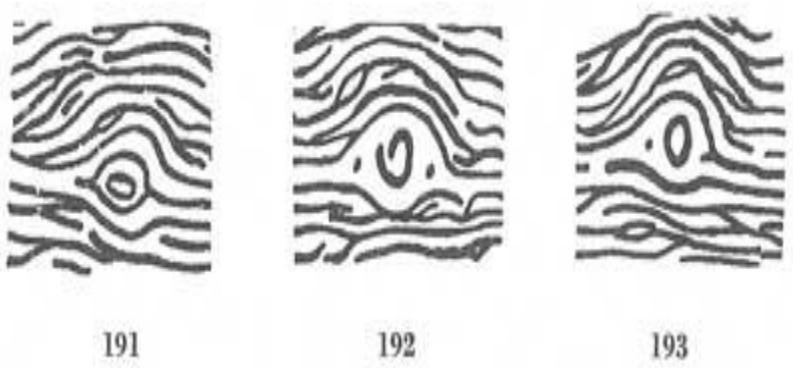
Type of LOOPs

- The terms **"radial"** and **"ulnar"** are derived from the radius and ulna bones of the forearm.
- Loops which flow in the direction of the ulna bone **(toward the little finger)** are called ulnar loops
- Loops which flow in the direction of the radius bone **(towards thumb)** are called radial loops.



Whorls

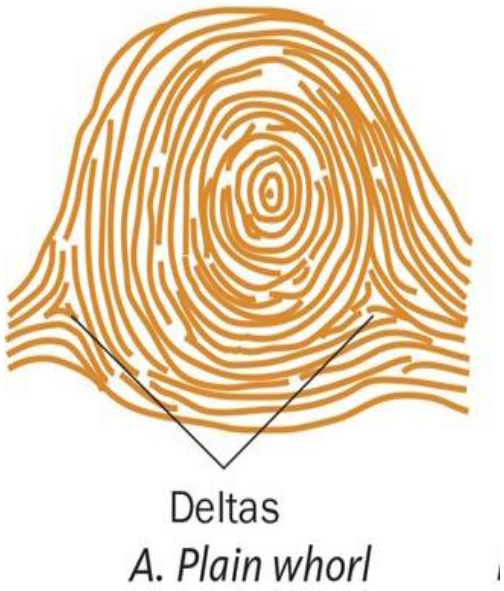
- The second most common pattern (30-35%).
- Two or more deltas (delta is the first ridge adjacent to the divergence point of two type lines) and there exists a recurve before reaching each delta.
- There are four sub-categories of whorls:
  - Plain Whorl,
  - Central Pocket Loop,
  - Double Loop,
  - Accidental.



191 192 193

Plain Whorl

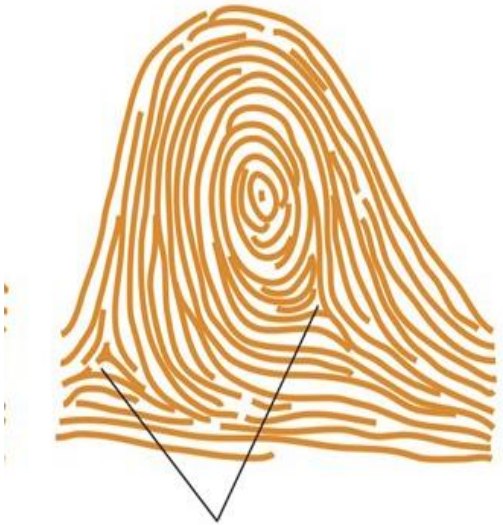
- A plain whorl pattern possesses type lines and at least two deltas.
- It also has at least one ridge that makes a complete trip.
- This ridge may be in the form of a spiral, oval, circle or variant of a circle.
- The plain whorl is the most common form of whorl and the mostly found.



A. Plain whorl

Central Pocket Loop

- In these whorls, one or more of the simple recurves of the plain whorl recurves a second time.
- An illusory line can be drawn amongst the two Deltas and does not cross or touch a ridge inside the type lines.
- One delta appears to be significantly adjacent to the center of the pattern than the other delta.



B. Central pocket loop whorl

Double Loop

- A Double Loop pattern, as the name denotes, is made up of two loops joint into one fingerprint.
- It comprises of two separate loop developments with two individual and different groups of shoulders and two deltas.



C. Double loop whorl

Accidental Whorl

- These are certain composite pattern inside the whorl group as they arise infrequently and their existence is purely by chance or by accident.

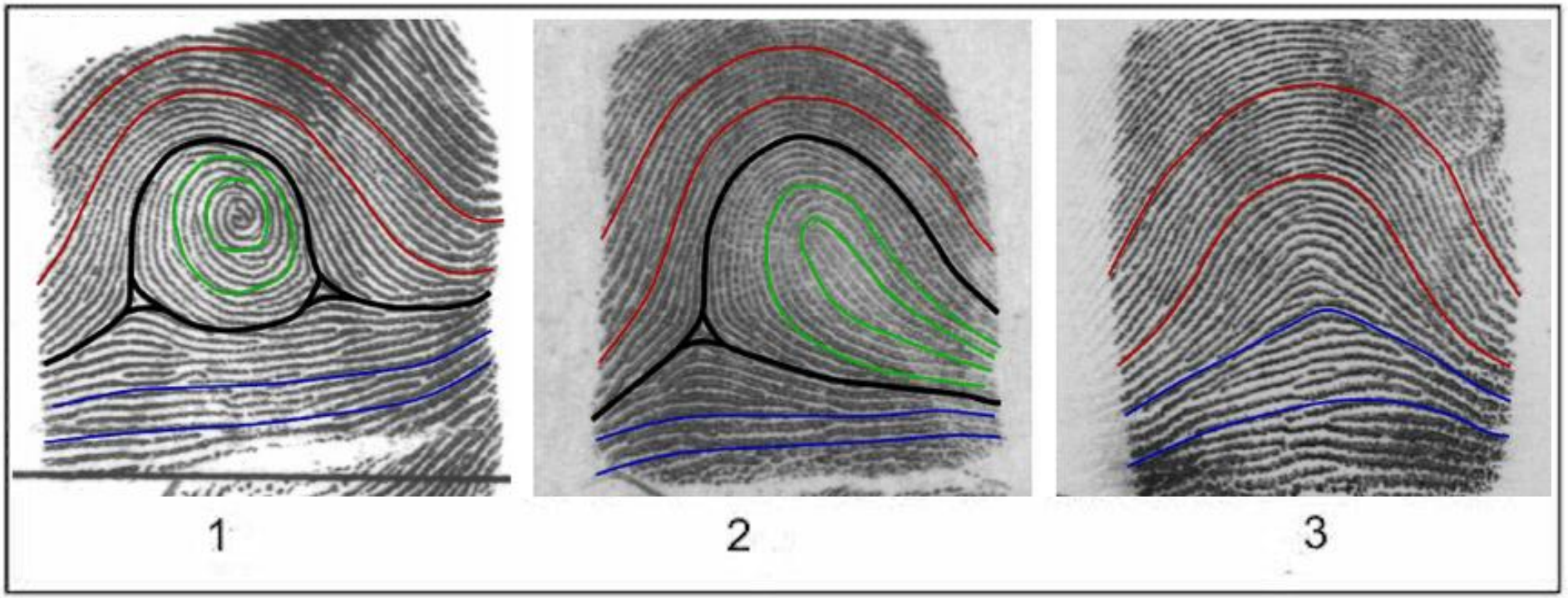
- In these, the combination of the pattern is derived from two diverse kinds of forms with at least two deltas.

  - Loop and a whorl
  - Loop and a tented arch
  - Loop and central pocket loop
  - Double loop and central pocket loop



Deltas

D. Accidental whorl


1
2
3

Significance of Fingerprints in Forensics

- Universally recognized, infallible means of personal identification.

1. Uniqueness
2. Infallibility/Reliable
3. Durability
4. Simplicity

Location of FPs

1. **The crime scene**

   Furniture, almirah, plastics, doors, glass, crockery, fridge, bottles, utensils, switches, sun-mica, photography paper, documents

2. **Point of entry**

   Locks, keys, broken glass, windowsills, frames

3. **The vehicle**

   Steering, wheels, mirrors, door handles, etc

4. **Culprit/objects related to culprit**

   Tools, weapons, cigarette, pan-masala pouch, disposable glasses, bottles, torch

5. **Human bodies/victims**

   Rape/strangulation cases, neck, wrists, breasts, thighs, ankles etc.

Finger Print Residue

- Complex mixture of secretions of SWEAT (Apocrine + Eccerine), SEBACEOUS glands.
- Very small amount (<10 µg) is deposited in very thin layer (0.1 µm).
- When deposited, 99% is water…which evaporates with time.

- Residue is-
  - **Water soluble**
    - ✓ Eccrine secretions
    - ✓ Salts (NaCl) and amino acids
    - ✓ Silver nitrates and ninhydrin are preferred for NaCl and AA
  - **Water insoluble**
    - ✓ Large proteins and lipids
    - ✓ Physical developer method

Sweat and Sebum Glands

- **Sweat Glands**
  - Apocrine glands
  - Eccrine glands

- Composed of
  - 99% water,
  - 1% organic and inorganic materials (chlorides, metal ions, lactic acid, ammonia, creatinine, and amino acids).

Sebaceous Glands

✓Associated with hair follicles,

✓Produce an oily, waxy substance called sebum that protects the hair and skin from water.

✓Secrete directly into the follicle rather than onto the skin.

✓Sebum largely consists of lipids (fats, oils, and waxes).

•Sebaceous secretions may be deposited on the fingertips through common actions such as running fingers through hair or scratching.

Sweat and Sebaceous Glands

Eccerine Glands

- Throughout the body (millions in number)
- Predominant in palms and soles
- 700-900gm/day
- 99% is water
- Amino acids, lipids, drugs, ethanol

Apocrine Glands

- Opens into hair follicle
- Predominant into armpits and pubic regions
- Secretions gets mixed with secretions of sebaceous glands before reaching skin surface
- Contains proteins, carbohydrates, cholesterol, iron, and steroids

Sebaceous Glands

- Sac like small organs
- Throughout the body
- Predominant in scalp, face, nose, mouth, anus etc.
- Absent in palms and soles
- Opens in hair follicles
- Fatty acids, oils, cholesterol, and waxes

Secretions of Eccrine Glands

Organic		Inorganic (major)		Inorganic (trace)
Amino acids	0.3–2.59 mg/L	Sodium	34–266 mEq/L	Magnesium
Proteins	15–25 mg/dL	Potassium	4.9–8.8 mEq/L	Zinc
Glucose	0.2–0.5 mg/dL	Calcium	3.4 mEq/L	Copper
Lactate	30–40 mM	Iron	1–70 mg/L	Cobalt
Urea	10–15 mM	Chloride	0.52–7 mg/L	Lead
Pyruvate	0.2–1.6 mM	Fluoride	0.2–1.18 mg/L	Manganese
Fatty acids	0.01–0.1 µg/mL	Bromide	0.2–0.5 mg/L	Molybdenum
Sterols	0.01–0.12 µg/mL	Iodide	5–12 µg/L	Tin
Creatine		Bicarbonate	15–20 mM	Mercury
Creatinine		Phosphate	10–17 mg/L	
Glycogen		Sulphate	7–190 mg/L	
Uric acid		Ammonia	0.5–8 mM	
Vitamins				

Secretions of Sebaceous Glands

Lipid class	Percentage
Triglycerides	30–50
Free fatty acids	15–30
Wax esters	12–16
Squalene	10–12
Monoglycerides and diglycerides	5–10
Cholesterol	1–3
Cholesteryl esters	1–3
Hydrocarbons	1–3

Significance of Fingerprint Residue

- **By understanding the residues that make up the material left behind on a surface as a fingerprint, a criminalist can target specific materials with the most effective technique for the circumstance to visualize the most prints.**
  - Black powder, for example, adheres to water and oils.
  - Silver nitrate reacts with chlorides.
  - Ninhydrin reacts with amino acids.

- The composition affects the **visibility**, **development technique**, and **persistence** of a fingerprint on different surfaces.

- Chemical composition helps select **appropriate reagents** for latent print development (e.g., ninhydrin for amino acids).

- Amino acids in the fingerprint residue have an affinity for the cellulose in paper and do not disperse over time.

- Because of this property, decades-old latent prints have been developed with ninhydrin.

Types of FP

- There are three types of fingerprints that can be found:
  - Latent
  - Patent

- **Latent fingerprints** are made of the sweat and oil on the skin's surface.
  - This type of fingerprint is invisible to the naked eye and requires additional processing in order to be seen. This processing can include basic powder techniques or the use of chemicals.

- **Patent fingerprints** can be made by blood, grease, ink, or dirt.
  - This type of fingerprint is easily visible to the human eye.

  - **Plastic fingerprints** are three-dimensional impressions and can be made by pressing your fingers in fresh paint, wax, soap, or tar.
  - Plastic fingerprints are easily seen by the human eye and do not require additional processing for visibility purposes.

Latent Fingerprints

- **LATENT PRINTS** - are fingerprint impressions secreted in a surface or an object and are usually invisible to the naked eye.
- These prints are the **result of perspiration** which is derived from **sweat pores found in the ridges** of fingers.
- When fingers touch other body parts, **moisture, oil and grease** adhere to the ridges so that when the fingers touch an object, such as a lamp, a film of these substances may be transferred to that object.
- The impression left on the object **leaves a distinct outline of the ridges** of that finger.
- These fingerprints **must be enhanced upon collection** and, because they serve as a means of identifying the source of the print.

Patent Fingerprints

- **PATENT PRINTS** - are visible prints that occur when a **foreign substance on the skin** of a finger comes in contact with the smooth surface of another object.

- These prints leave a **distinct ridge impression** that is **visible with the naked eye** without technological enhancement of any kind.

- The tried and true "blood on his hands" evidence is an example of patent prints recovered from a crime scene or scene of interest to investigators.

- These foreign substances contain dust particles which adhere to the ridges of the fingers and are easily identifiable when left on an object.

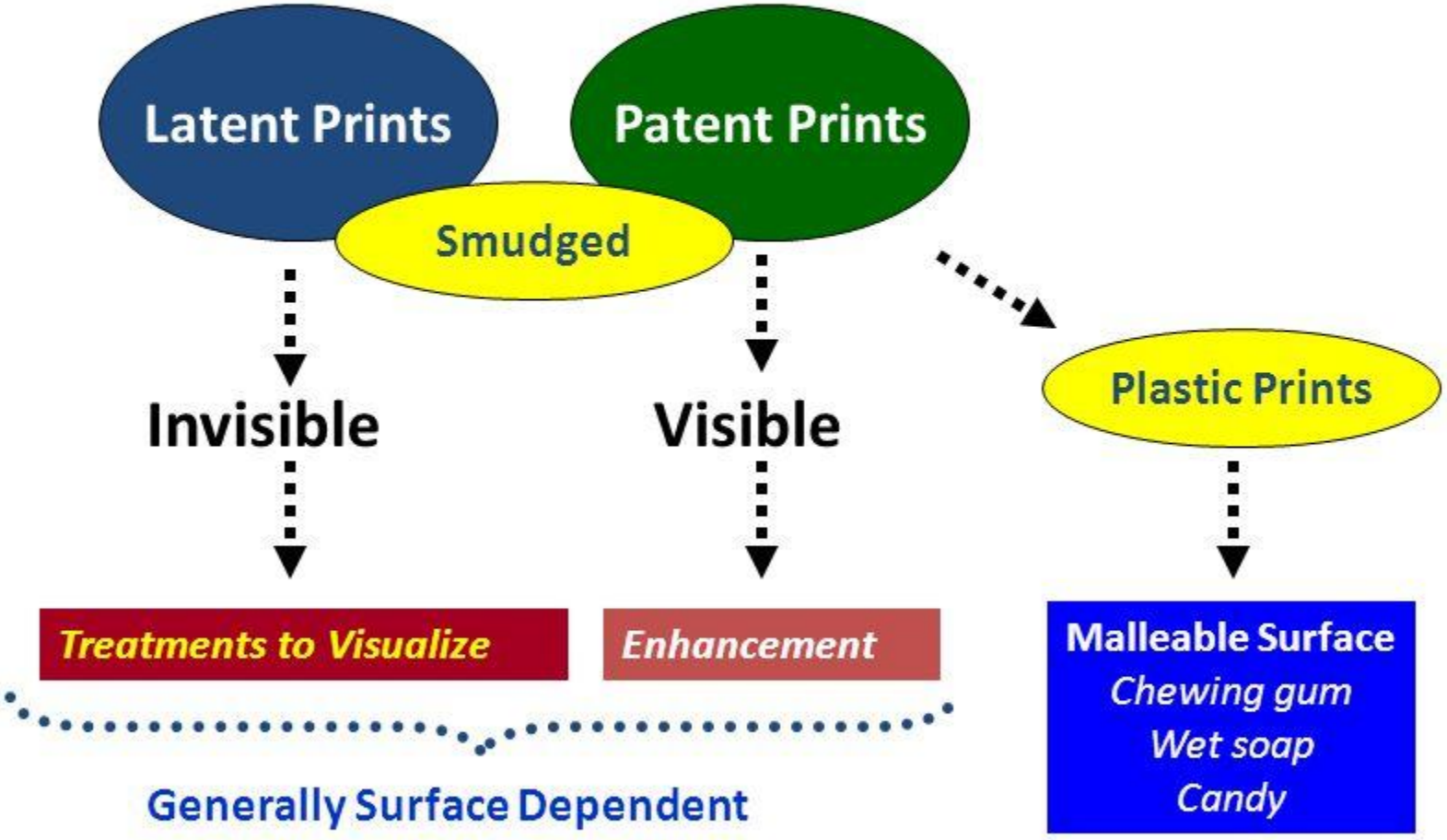
Plastic Fingerprints

- **PLASTIC PRINTS** - are **visible**, **impressed prints** that occur when a finger touches a soft, malleable surface resulting in an indentation.

- Some surfaces like freshly painted or coated, or those that contain **wax, gum, blood** or any other substance that will soften when hand held and then retain the finger ridge impressions.

- These prints require no enhancement in order to be viewed, because they are impressed onto an object and **are easily observable**.

Types of Fingerprints



Types of Surfaces/Substrates for FP Deposition

- The general characteristics of the surface are:
  - **Porous** (paper, cardboard, and untreated wood),
  - **Non-porous smooth** (varnished or painted surfaces, plastics, and glass),
  - **Non-porous rough** (leather, and other textured surfaces).

- The distinction between porous and non-porous surfaces is their ability to absorb liquids. Liquids sink in when dropped onto a porous surface, while they sit on top of a non-porous surface.

  - For porous surfaces, scientists sprinkle chemicals such as ninhydrin over the prints and then take photographs of the developing fingerprints.

  - For non-porous smooth surfaces, experts use powder-and-brush techniques, followed by lifting tape.

  - For rough surfaces, the same powdering process is used, but instead of using regular lifting tape for these prints, scientists use something that will get into the grooves of the surface such as a gel-lifter or Mikrosil (a silicone casting material).

Porous Surfaces

- Porous substrates are generally **absorbent** and include materials like
  - **paper,**
  - **cardboard,**
  - **wood**, and
  - **other forms of cellulose**.

- Fingerprints deposited onto these media are **absorbed into the substrate** and are somewhat **durable**.

- **Amino acid techniques** are particularly useful here because the amino acids tend to remain stationary when absorbed and do not migrate.

Non-Porous Surface

- Nonporous surfaces **do not absorb**.
- These surfaces **repel moisture** and often **appear polished**.
  - glass,
  - metal,
  - plastics,
  - lacquered or painted wood,
  - rubber.
- Latent prints on these substrates are **more susceptible to damage** because the fingerprint residue resides on the outermost surface.
- **Cyanoacrylate (CA), dye stains, powders**, and **vacuum metal deposition** are usually the best choices to use on these surfaces.

Factors Affecting the FP deposition/Transfer

Pre-transfer Factors

- Condition/**health** of the donor's friction skin (age, gender, stimuli, occupation, disease)
- Amount and **type of residue** on the skin (disease, and any substances the subject may have touched prior to deposition)

Transfer Factors

- Conditions of the **surface/substrate** being touched (texture, surface area, surface curvature or shape, surface temperature, condensation, contaminants, and surface residues).
- The **pressure applied** during contact (deposition pressure, lateral force)

Post-transfer Factors

- **Environmental factors** (water, humidity, and temperature)
- Physical contact from **another surface**

Factors Influencing the Choice of Selection of Development Method

- Type of latent print residue suspected
- Type of substrate
- Texture of substrate
- Condition of substrate (clean, dirty, tacky, sticky, greasy, etc.)
- Environmental conditions during and following latent print deposition
- Length of time since evidence was touched
- Consequences of destructive processing methods
- Subsequent forensic examinations
- Sequential ordering of reagents

Powdering Methods

- *Latent print visualization with powder, or "dusting", involves the application of finely divided particles that physically adhere to the aqueous components in latent print residue on nonporous surfaces.*

- The **oldest** and **most common** methods of latent print detection.

- Various locally available substances like **charcoal**, **lead powder**, **cigar ashes**, **powdered iron**, **soot**, and **talc** etc have been used.

- Latent print powder particles have an **affinity for moisture** and preferentially clings to the residue deposited by friction ridge skin.

Powdering Methods

- The powder technique for detecting latent fingerprints involves the **application of a finely divided formulation** to the fingermark impression, generally **with a glass-fibre or a camel hair brush**.
  - The powder gets mechanically adhered to the **sweat residue** defining the ridge pattern.
  - The **furrows which are devoid of the fingerprint residue** do not adhere the power onto them.
  - The final outcome is that the **powder formulation sticks to the ridges, but is easily blown off the furrows.**
  - Since the powder is normally colored, the ridge pattern becomes visible and the **latent print gets developed/visible**.

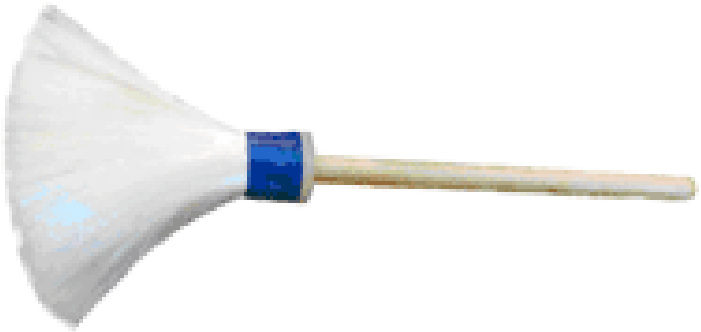
Steps in Brushing Method

- Brushes should be clean, dry, and relatively free of tangles.
- **"Loading" The Brush (**conventional brush**):**
  - The filament tips are lightly dipped into a sterile, wide-mouth container holding a small amount of powder.
  - Excess powder is then shaken, spun, or tapped from the brush.
- The powder is then **applied evenly to all areas** of the substrate.
- Brushing is accomplished with **light and even strokes.**
- After the appearance of latent prints, the excess powder can be subsequently **brushed away along the direction of ridge flow**.
- **Lifting the developed prints** using transparent tape and storing in file/fingerprint card.

Precautions

- Powders are typically applied **to nonporous surfaces** with a soft brush.
- The **softness of the bristles** is particularly important to prevent damage to fragile latent print residue.
- Conventional brushes are typically made with **animal hair, fiberglass filaments**, or sometimes **feathers**.
- Latent **prints with a high moisture or oil content are easily damaged** by a brush that is too stiff or is used with excessive force.
- Powders applied with a traditional filament brush consist of **very fine particles** and are usually **low density** or “fluffy” in nature. This enables particles to be easily picked up or “loaded” onto the brush filaments.
- The effective adhesion of powder depends on the size and shape of the particles that compose the formulation. **Small, fine particles adhere more easily** than large, coarse ones.


"HI-FI" LATENT
PRINT POWDER



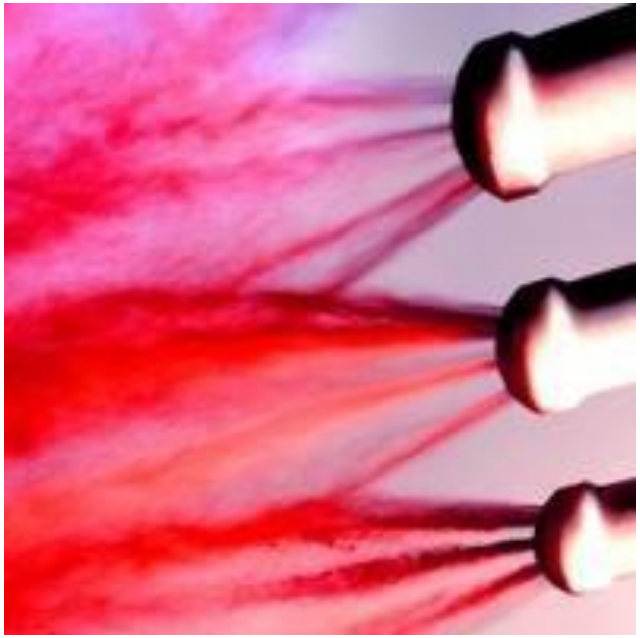




Electrostatic Spraying

•10% of all fingerprints developed using powdering by brushing at crime scene results in difficulty in identification .

•**Electrostatic depositor** or **atomizer** or **aerosol spray** methods are used to avoid such damages to latent prints.



- The power is placed on the surface of an electrode held about 2.5 cm above the surface with latent fingerprint.
- A potential gradient of about 12 kV is maintained between the electrode and the surface.
- The powder on the electrode becomes charged to the polarity of the electrode and is accelerated towards the surface, where it is discharged.

Magnetic Powders

- Magna-powders are made by incorporating coarse, spherical iron particles in the conventional powders.
- The magnetic iron acts as a carrier for the non-magnetic formulation, forming a brush when the powder is picked up with a magnetized applicator.
- On brushing the imprint impression, only the fine particles of the formulation adhere to the fingerprint residue.
- Once the print has been developed, the excess powder can be detached from the applicator by withdrawing the magnetized steel rod.

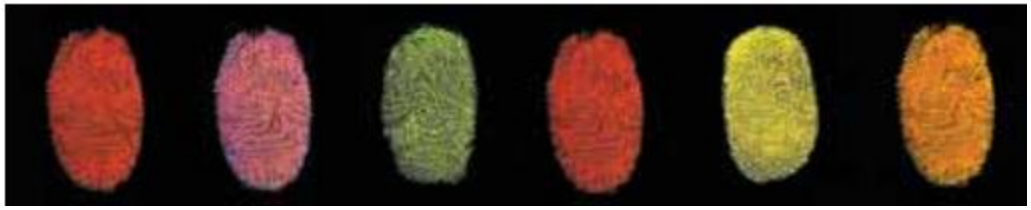


Fingerprint Powders

- **Pigment**
  - Provides for effective visualization,
  - offers contrast and definition against the background surface,
  - Some pigment powders offer enough adhesion to be used individually without binder
  - carbon black (colloidal carbon), lamp black, talc, kaolin, aluminum, metal flake, and dolomite

- **Binder**
  - Provides for maximum and preferential adhesion to latent print residue
  - Also referred to as the carrier
  - Iron powder, lycopodium, corn starch, rosin, and gum arabic

Organic Dye Based (Fluorescent) Powders



- Commercial fingerprint powers containing inorganic salts of **mercury, cadmium, titanium, lead, manganese**, etc have been replaced with the organic-based formulations.

- These formulations generally contain an organic dye as the colorant.
- For improved results fluorescent and laser-active dyes are used –
  - **Fluorescein,**
  - **Eosin Y**
  - **Rhodamine B**
- Useful for visualization of weak latent prints and prints deposited on multi-colored surfaces that would present a contrast problem

Use FLUORESCENT powders with:

1. Forensic Light Sources
2. Lasers (Argon Ion, Copper Vapor, and Nd (neodymium) Yag Laser)
3. Ultraviolet Lights (Longwave and shortwave)
4. Bluemaxx (455nm) Alternate Light Source w/orange filter

Ninhydrin

- Ninhydrin was first described in 1910 when **Siegfried Ruhemann.**
- Reacts **with Amino Acids** (AA) and gives **purple** color.
- Wide applications started immediately in analytical chemistry **for detecting proteins/AAs**.
- Most popular methods for fingerprint detection on paper and other **porous substrates.**

Physical properties of Ninhydrin

Ninhydrin is

- Colorless
- Crystalline solid
- Highly soluble in polar solvents such as water and methanol
- When heated, the solid becomes pink to red in color at approximately 125 °C
- Melts at 130–140 °C

Principle of Ninhydrin FP Development

- Aqueous deposits of sweat (99%) contain a small, but detectable amount of **amino acids (avg 250ng)** per fingerprint.
- After the **water evaporates** from the surface, the amino acids remain as solid material.
- AAs have a **strong affinity to cellulose** of paper and doesn't migrate much with time.
- AAs give **sharp purple color** with Ninhydrin.
- The affinity of **Ninhydrin is not specific to certain AA**, but gives color with all 14 AAs present in sweat.

Ninhydrin Developed FPs

Prepare Ninhydrin solution in water

↓

Spraying

↓

Air drying

↓

Heating

↓

$ZnCl_2$

↓

Photograph/ document

- Ruhemann's purple is known to **degrade in the presence of light and oxygen,** the treated fingerprint should be stored in a dark, cool place.
- Provides excellent contrast under ideal conditions (e.g., **fresh fingerprints on white paper**).
- On colored paper or with aged fingerprints, however, the FP development is weaker.
- Addition of **salts of Zinc**, Cadmium, Cobalt, and Copper can preserve Ninhydrin spots and **provides a contrast on colored background.**

DFO Method

- **1,8-Diazafluoren-9-one (DFO)** was used in 1990 as fingerprint development reagent.
- DFO reacts with **AAs** on porous surfaces and gives **pinkish red** color.
- Faint red or pink colored fingerprints appeared and were **intensely fluorescent at room temperature.**
- Presented clear **advantages over the metal complexation induced fluorescence of ninhydrin-developed fingerprints**.
- DFO is reported to be a **more sensitive** fingerprint development reagent than ninhydrin, producing a **greater number of identifiable latent fingerprints.**
- **Combination of DFO followed by ninhydrin** develops more latent fingerprints than DFO or ninhydrin alone.

DFO Development Steps

Prepare DFO solution

↓

Spraying

↓

Air dried

↓

Heated (100 °C oven for 10–20 minutes)

↓

Documentation/photography (using 530nm excitation light and a 590nm barrier filter, or 555nm excitation light and a 610nm barrier filter









DFO > Ninhydrin

- With DFO, a strong room temperature luminescence of developed prints is observed **without any secondary treatment**.
  - Against this, the ninhydrin developed prints have to be post-treated with a suitable metal salt to produce luminescence. Good results are obtained if the exhibit is cooled to -200° C during the fluorescence examination.

- With DFO reagent, results are obtained in a **relatively short time of 30 minutes**.
  - Ninhydrin, on the other hand develops fingerprints at room temperature in about 2 hours. For faster development, the surface in question has to be heated to about 80°C. For improved results, heating should be carried out in a humid atmosphere.

- On an average, **DFO reagent develops 3 times more fingerprints** than ninhydrin.

- If DFO reagent fails to detect a latent impression, **ninhydrin method can still be tried out on the same surface**.
  - However, if ninhydrin reagent does not develop a particular imprint, DFO method cannot be extended for its detection

Demerits of DFO over Ninhydrin method

- A high-powered light source, with **appropriate filter, must be available for observing luminescence** of DFO-developed fingerprints. Otherwise, weak impressions will not be detected.
- If the surface on which latent fingerprints are to be developed shows **background fluorescence**, the utility of the DFO method is curtailed.
- DFO reagent **does not give satisfactory results on heat sensitive items like thermal paper**, window envelops and labels on polyethylene articles.
- DFO reagent is **costlier than ninhydrin**.
- Working solution of **DFO reagent is not very stable**.

1,2-Indanedione

- 1,2-Indanedione is a **close analogue of Ninhydrin**
- Reacts with amino acids in a very similar fashion.
- Results of FP development are:
  - **Indanedione-DFO sequence > 1,2-Indanedione > DFO–Ninhydrin sequence > DFO > Ninhydrin**
- Other Ninhydrin analogues is **5-Methylthioninhydrin (5-MTN)**

Lipid-Sensitive Reagents

- Oil Red O and Nile Red are Lipid-Sensitive Reagents commonly used in development of latent fingerprints based on their interaction with lipids (fats and oils) present in fingerprint residue.

- These reagents are particularly useful when dealing with porous surfaces where water-soluble components of the fingerprint (like amino acids and salts) are absorbed into the substrate, but the lipid-based materials remain on the surface.

Oil Red O

- **Oil Red O (ORO)** is a fat-soluble dye that binds to the lipophilic (oil-loving) components of fingerprint residues.

- **Mechanism of Action:**
  - Oil Red O selectively stains the **lipid components** in the fingerprint, such as sebaceous oils, fats, and lipids.
  - When applied to a surface with latent fingerprints, the dye binds to the oils in the ridges of the print, producing a visible contrast between the stained ridges and the non-stained background.

- **Steps in Using Oil Red O:**
  1. **Application**: The surface with the suspected fingerprint is immersed in or sprayed with a solution of Oil Red O in a mixture of solvents like methanol and water.
  2. **Rinsing**: The surface is rinsed to remove excess dye that has not adhered to the fingerprint.
  3. **Development**: The stained lipids in the fingerprint ridges become visible as bright red lines, allowing the print to be photographed or analyzed.

Oil Red O

- **Advantages**
  - **Effective on porous surfaces**: Oil Red O works well on surfaces like paper or cardboard where traditional fingerprinting methods may fail.
  - **Detects older fingerprints**: Since it targets lipids, it can reveal fingerprints that are weeks to months old, even after water-soluble components have degraded.
  - **Non-destructive**: It can be used before applying other fingerprinting techniques without damaging the print.

- **Disadvantages**
  - **Limited to lipid-rich prints**: Oil Red O is only effective on fingerprints that contain sufficient amounts of lipids. If a fingerprint contains mainly water-soluble components, it may not develop well.
  - **Sensitivity to background**: On some surfaces, the dye may stain the background, making it harder to distinguish the print.
  - **Color limitations**: The red color may be hard to detect on certain backgrounds unless viewed under optimal conditions.

Nile Red

- **Nile Red** is another lipid-sensitive reagent that binds to fats and oils in latent fingerprint residue.
- Unlike Oil Red O, Nile Red is **fluorescent**, meaning it emits light when exposed to certain wavelengths, which makes it especially useful for developing prints in situations where contrast between the fingerprint and the surface is low.

- **Mechanism of Action:**
  - Nile Red interacts with the lipid components in the fingerprint and fluoresces under **UV or blue light**, producing a visible image of the fingerprint ridges.
  - Its fluorescence properties depend on the environment in which it is used. For example, in a hydrophobic (water-repelling) environment, Nile Red may emit different colors (such as yellow, red, or orange) when exposed to specific light wavelengths.

- **Steps in Using Nile Red:**
  1. **Application**: Nile Red is applied to the surface via a solution in solvents (such as ethanol or acetone).
  2. **Illumination**: After the application, the surface is illuminated with UV or blue light.
  3. **Observation**: The lipid-rich ridges of the latent print fluoresce, allowing them to be visualized and documented.

Nile Red

- **Advantages**
  - **Fluorescent**: The fluorescence enhances visibility, especially on surfaces where traditional development methods might not produce enough contrast (e.g., dark or multicolored backgrounds).
  - **High sensitivity**: It is particularly effective at detecting even very faint lipid-based prints.
  - **Non-destructive**: Nile Red can be used in combination with other techniques, such as powdering, without damaging the print.

- **Disadvantages**
  - **Fluorescence equipment required**: A major limitation of Nile Red is that it requires a fluorescence microscope or UV/blue light sources for the developed fingerprints to be visible.
  - **Background fluorescence**: Some materials may also fluoresce under UV light, which can make distinguishing the fingerprint from the background difficult.
  - **Special handling**: Fluorescent reagents like Nile Red often require careful handling and specific equipment for proper visualization.

Oil Red O vs Nile Red

ASPECT	OIL RED O	NILE RED
Detection Method	Visible red stain	Fluorescence (under UV or blue light)
Sensitivity to Lipids	Good sensitivity to lipids	Highly sensitive to lipid components
Surface Type	Primarily used on porous surfaces (e.g., paper)	Can be used on both porous and non-porous surfaces
Visibility/Contrast	Red color on most surfaces, but may have low contrast on certain backgrounds	High contrast due to fluorescence, especially on dark or multicolored surfaces
Equipment Required	Minimal – visible light	UV/fluorescent light source or microscope required
Advantages	Simple and effective, detects older prints	Highly sensitive, produces clear fluorescent prints
Disadvantages	May not develop lipid-poor prints, color limitations	Requires special equipment, background fluorescence can interfere
Common Uses	For porous surfaces like paper and documents	For faint prints, dark or multi-colored backgrounds

Iodine Fuming Method

- Iodine is a crystalline solid, which upon **heating sublimes into violet vapors**.
- Reacts with **fatty acids** of sweat deposition

- Methodology
  - Fuming Gun Method
  - Fuming Cabinet Method
  - Powder Dusting Method

- **Fades quickly**
- Should be **photographed immediately**
- **Iodine-starch** can give permanent blue color
- Post-treatment with **N,N,N',N'-tetramethyl-4,4'-diaminodiphenylmethane in 1,2,2-tricholorotrifluoromethane** gives green blue color.
- Post-treatment with **Phenothiazine** gives red color.
- **7,8-benzoflavone-Iodine fuming** is most suitable, non-destructive method for post-treatment of old latent prints from old documents.

Cyanoacrylate Fuming

- Super glue
- Virtually all **nonporous surfaces-**
  - Glass,
  - Metal,
  - Coated papers,
  - Plastics
- Fully developed CA prints are a white **3-D** matrix, often visible to naked eyes
- Polymerization of **Cyanoacrylate ester monomer**



Guide to Surfaces

Surface	Treatment Method
Smooth, non-porous: glass, hard plastic mouldings, surfaces treated with paint or varnish	Powders, iodine, small particle reagent, cyanoacrylate/ fluorescent dyes
Rough, non-porous: rough, textured surfaces, grained plastic mouldings	Small particle reagent, cyanoacrylate/fluorescent dyes
Paper & cardboard: paper & cardboard that hasn't been waxed or plastic-coated.	Iodine, ninhydrin, DFO, silver nitrate, or physical developer. Powders won't work with older fingerprints
Plastic packaging media: Polyethylene, polypropylene, cellulose acetate, and laminated paper surfaces	Iodine, small particle reagent, cyanoacrylate/fluorescent dyes, and powders
Soft vinyl (PVC), rubber and leather: simulated leather and cling film	Iodine, small particle reagent, cyanoacrylate, and powders
Untreated metal: untreated, bare metal surfaces; metal surfaces that haven't been painted or lacquered.	Small particle reagent, powders, cyanoacrylate / fluorescent dyes and powders
Unfinished wood: wood surfaces that haven't been painted or treated.	Ninhydrin. You can use powders on smooth wood and silver nitrate or physical developer on light woods.
Wax and waxed surfaces: items made of wax, such as candles, wax-coated paper, cardboard, wood surfaces	Nonmetallic powders, cyanoacrylate / fluorescent dyes
Adhesive-coated surfaces: tapes and similar surfaces that are not likely to dissolve in water	Adhesive-side powders